MILITARY SPECIFICATION

CARTRIDGE, 81MM, SMOKE, WP, M375A3 ASSEMBLING, MARKING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Specification covers the assembling, marking and packing for one type of smoke mortar cartridge designated as Cartridge, 81MM, Smoke, WP, M375A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-48156 - Cartridge, Ignition, XM299, LAP
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

9294735 - Cartridge, 81MM, Smoke, WP, M375A3, W/Fuze, PD, M567
9230175 - Container, Ammo, Fiber, M252A4, for 81MM Mortar Ammunition in Jungle Wrap

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9278234 - Box, Wirebound, Packing, Ammo for 81MM Mortar
9287601 - Packing and Marking for Box

INSPECTION EQUIPMENT DRAWINGS

6518071 - Profile and Alignment gage
6518072 - Alignment Check gage
7257965 - Wear limit check gage

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and Assemblies.-The components and assemblies shall comply with all requirements specified on drawing 9294735 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Proving Ground Test.-The cartridges shall comply with the following requirements.

3.3.1 Misfires.-The cartridges shall not fail to fire from the weapon.

3.3.2 Duds.-The cartridges shall function on impact. There shall be visible evidence of functioning as indicated by presence of a smoke cloud.

3.3.3 Short Rounds.-No cartridge shall have a horzontal range less than eighty (80) percent of the average horizontal range (calculated without the short rounds).

3.3.4 Premature Burst.-There shall be no premature burst in the gun bore or in flight.

3.3.5 Metal Parts Separation.-There shall be no metal parts separation in the gun bore or in flight.

3.3.6 Range Standard Deviation.-The horizontal range standard deviation shall be no greater than twenty (20) meters at charge zero and no greater than fifty (50) meters at charge four.

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3.4 Workmanship.-Parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-The provisions of MIL-A-48078 shall apply.

4.2 Quality Conformance Inspection

4.2.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of Cartridges, 81MM, Smoke, WP, M375A3 shall contain:

a. WP loaded filling assemblies from one manufacturer in accordance with the same drawing, same drawing revision, same specification and specification revision of one loading lot interfix number.
b. Ignition cartridges, of one lot interfix number, from one manufacturer.
c. Fin assemblies, of one lot interfix number, from one manufacturer.
d. Propelling charges, made from not more than one lot of propellant, from one manufacturer.
e. Loaded fuzes of one lot interfix number from one manufacturer.
f. Loaded projectile bursters, of one lot interfix number, from one manufacturer.
g. Obturating rings, of one lot interfix number, from one manufacturer.

4.2.2 Examination.-See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.1	Body Filling Assembly				8835263
					NEXT HIGHER ASSEMBLY 9294735
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Evidence of WP leakage		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.2	Cartridge (prior to inserting burster assembly)				9294735
					NEXT HIGHER ASSEMBLY 9294735
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Burster pad(s) missing		100%	3.2	Visual
Major A					
101	Foreign matter in burster cavity		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9294735
4.2.2.3	Cartridge (prior to assembly of fuze)				NEXT HIGHER ASSEMBLY 9294735
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major A					
101	Burster charge missing		0.40%	3.2	Visual
102	Foreign matter in fuze well		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9294735
4.2.2.4	Cartridge				NEXT HIGHER ASSEMBLY 9294735
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Ignition cartridge head not fully seated		100%	3.2	Visual
2	Fuze safety wire missing or insecure		100%	3.2	Visual-Manual
3	Propelling charges missing or damaged (cracked, delaminated)		100%	3.2	Visual
4	Obturating ring missing		100%	3.2	Visual
5	Fin assembly cracked, damaged or distorted		100%	3.2	Visual
Major A					
101	Obturating ring not welded		100%	3.2	Visual
102	Obturating ring not flush or below bourrelet diameter		100%	3.2	Visual
103	Cartridge fails to freely enter profile and alignment gage		100%	3.2	Gage/Dwg. 6518071 Gage/Dwg. 6518072 Gage/Dwg. 7257965

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.2.2.4	Cartridge				9294735
					NEXT HIGHER ASSEMBLY 9294735
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major A					
104	Obturating ring damaged		0.40%	3.2	Visual
105	Fin assembly fails to meet minimum assembly torque		0.40%	3.2	Test
106	Fuze fails to meet minimum assembly torque		0.40%	3.2	Test
107	Ignition cartridge fails to meet minimum assembly torque		0.40%	3.2	Test
108	Fin assembly not fully seated		0.40%	3.2	Visual
109	Fuze not fully seated		0.40%	3.2	Visual
Minor					
201	Paint damage area in excess of 1/4 square inch		0.65%	3.2	Visual
202	Pettman cement does not cover a minimum of one half the number of threads for a full 360 degrees (applicable to the fuze, fin and ignition cartridge - prior to assembly)		0.65%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.2.2.4	Cartridge				9294735
					NEXT HIGHER ASSEMBLY
					9294735
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Marking misleading or unidentifiable		0.65%	3.2	Visual
204	Warning labels missing		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.4	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9230175
4.2.2.5	Unsealed fiber container				NEXT HIGHER ASSEMBLY 9230175
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Packing stop missing		0.65%	3.2	Visual
202	Spacer, cushion or filler disc assembly missing		0.65%	3.2	Visual
203	Propelling charge supports bind on inside diameter of container preventing easy removal of round		0.65%	3.2	Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9230175
4.2.2.6	Sealed fiber container				NEXT HIGHER ASSEMBLY 9230175
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major A					
101	Container damaged		0.40%	3.2	Visual
102	Sealing strip incomplete or badly wrinkled		0.40%	3.2	Visual
103	Metal end loose or distorted		0.40%	3.2	Visual/Manual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.7	Fiber container in jungle wrap				9230175
					NEXT HIGHER ASSEMBLY
					9230175
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major B					
131	Wax coating incomplete		0.40%	3.2	Visual
132	Inner wrapper ends improperly folded, exposing fiber container		0.40%	3.2	Visual
Minor					
201	Length of sealing strip tab		0.65%	3.2	Gage
202	Marking missing or unidentifiable		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48394 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9278234
4.2.2.8	Sealed wirebound box				NEXT HIGHER ASSEMBLY 9278234
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Boards split or separated		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	Marking misleading or unidentifiable		0.65%	3.2	Visual
204	Strap insecure		0.65%	3.2	Manual/Visual
205	Wire broken		0.65%	3.2	Visual
206	Wireloops improperly fastened		0.65%	3.2	Visual

NOTES:

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4.2.3 Testing

4.2.3.1 Check Test for Possible Deterioration of Cartridge, Ignition, M299 (see Specification MIL-C-48156).-If the total elapsed time between original acceptance of the Ignition Cartridge lot and the assembly of that lot into complete round cartridges exceeds two years, or if the Ignition Cartridges have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-C-48156 immediately before the Ignition Cartridge lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges.

4.2.3.2 Strength of Heat Weld (see dwg. 9294735) Major defect.-One obturating ring sample, minimum, after every 350 cartridges, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.3.2. Failure of the sample to meet the requirement shall be cause for those 350 preceding cartridges represented by the sample to be rebanded and retested.

4.2.3.3 Proving Ground Test.-The cartridges shall be observed for the following characteristics. Testing shall be in accordance with 4.3.1:

a. Misfires, major defect (see 3.3.1).
b. Duds, major defect (see 3.3.2).
c. Short rounds, critical defect (see 3.3.3)
d. Premature burst, critical defect (see 3.3.4)
e. Metal parts separation, critical defect (see 3.3.5)
f. Range standard deviation, major defect (see 3.3.6)

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, eighty random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. two or more misfires occur
b. four or more duds occur
c. one or more short rounds occur
d. one or more premature bursts occur

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e. one or more metal parts separations occur
f. horizontal range standard deviation at charge 0 or charge 4, multiplied by the factor 0.79 exceeds the applicable requirement.

4.2.3.3.2 After three consecutive lots have complied with the criteria of 4.2.3.3.1, forty eight random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. two or more misfires occur
b. four or more duds occur
c. one or more short rounds occur
d. one or more premature bursts occur
e. one or more metal parts separations occur
f. horizontal range standard deviation at charge 0 or charge 4, multiplied by the factor 0.75 exceeds the applicable requirement.

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.2.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Section 6 herein.

4.3 Test Methods and Procedures

4.3.1 Proving Ground Test.-The test shall be performed at a Government Proving Ground in accordance with the applicable USA-TECOM Acceptance Test Procedure.

The cartridges shall be fired from a standard 81mm mortar at an elevation of forty-five degrees. One half of the sample cartridges shall be fired at charge one and the other half at charge nine. All fuzes shall be set for superquick functioning. The individual velocities, pressures and ranges shall be recorded on the firing record. Observation shall be made for misfires and duds, and shall be recorded. In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal. Misfired ignition cartridges shall not be disassembled by the Proving Ground. Should a permature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard

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deviations and arithmetic averages shall be calculated and recorded. Range probable error shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of those rounds, and prints of the preceeding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded. Smear camera shall be used to photograph all samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in that particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded.

4.3.1.1 Test Validity.-If for any reason the proving ground or Picatinny Arsenal considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

4.3.2 Strength of Heat Weld.-The sample shall be tested for integrity by applying the required load at the heat weld radially and in a direction away from the axis (center) or the ring.

5 PREPARATION FOR DELIVERY

5.1 Level of Packaging and Packing.-Packaging and packing shall conform to Level A, as established in DOD Instructions 4100.14 "Uniform Preservation - Packaging, Packing and Marking of Items of Supply".

5.2 Packaging, Packing and Marking.-Packaging, packing and marking shall be in accordance with dwg. 9230175 and dwg. 9287601.

6. NOTES

6.1 Ordering Data.-See MIL-A-48078 (PA).

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6.2 Submission of Inspection Equipment Designs for Approval.- See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.3 Proving Ground Test Summary.

TEST	SAMPLE SIZE	REQUIREMENTS
Ballistic	(see 4.2.3.3) Eighty samples until three consecutive lots have complied with the acceptance criteria. Forty-eight samples thereafter.	a. Misfires (see 3.3.1) b. Duds (see 3.3.2) c. Short rounds (see 3.3.3) d. Premature Burst (see 3.3.4) e. Metal Parts Separation (see 3.3.5) f. Range Standard Deviation (see 3.3.6)

6.4 Submission of Test Data.-One copy of proving ground ballistic acceptance test data (i.e. firing record) will be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A, SARPA-AD-E-A3
Dover, New Jersey 07801

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1315-A-058

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OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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